H-B-420K
August 30, 1991
TO SUPERSEDE
H-B-420J (Dec. 30, 1985)
H-B-695H (Oct. 24, 1985)

FEDERAL SPECIFICATION

BRUSH, PAINT

This specification is approved by the Assistant Administrator, Office of Federal Supply Service, General Services Administration for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION.

1.1 SCOPE. This specification covers flat, metal-bound, paint and varnish brushes made with natural hog bristle or synthetic filament. Master grade and High grade brushes are designed and manufactured for paint applications where superior paint loading and release, and minimal brush marks are required. Grades B and C brushes deliver excellent results when performing utility painting operations.

1.2 Classification. The brushes shall be of the following types, grades, and sizes as specified (see 6.2): (Size denotes the width of brush inside the ferrule.)

	TYPE I. BRISTLE Size, inches (mm.)		TYPE II. FILAMENT Size, inches (mm.)	
Grade AA, Master grade	6	(152)	5	(127)
	5	(127)	4	(102)
	4	(102)	3-1/2	(89)
	3-1/2	(89)	3	(76)
Grade A, High grade	6	(152)	6	(152)
	5	(127)	5	(127)
	4	(102)	4	(102)
	3-1/2	(89)	3-1/2	(89)
	3	(76)	3	(76)
	2-1/2	(64)	2-1/2	(64)
	2	(51)	2	(51)
	1-1/2	(38)	1-1/2	(38)
	1	(25)	1	(25)
Grade B, Medium grade	4	(102)	5	(127)
	3-1/2	(89)	4	(102)
	3	(76)	3-1/2	(89)

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	TYPE I. BRISTLE Size, inches (mm.)		TYPE II. FILAMENT Size, inches (mm.)	
Grade B, Medium grade	2-1/2	(64)	3	(76)
	2	(51)	2-1/2	(64)
	1-1/2	(38)	2	(51)
	1	(25)	1-1/2	(38)
	---		1	(25)
Grade C, Utility grade	4	(102)	4	(102)
	3-1/2	(89)	3-1/2	(89)
	3	(76)	3	(76)
	2-1/2	(64)	2-1/2	(64)
	2	(51)	2	(51)
	1-1/2	(38)	1-1/2	(38)
	1	(25)	1	(25)
	1/2	(13)	---	

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Commercial Item Descriptions (CID):

A-A-857 -Thinner, Dope and Lacquer (Cellulose Nitrate)

Federal Specifications:

TT-B-838 -Butyl Acetate; Normal (For use in Organic Coatings)
TT-M-261 -Methyl Ethyl Ketone (For use in Organic Coatings)
PPP-B-1880 -Brushes; Preparation for Delivery of
TT-T-291 -Thinner, Paint, Mineral Spirits, Regular and Odorless

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks and Commercial Item Descriptions as outlined under General Information in the Index of Federal specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Auburn, WA.)

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 -Sampling Procedures and Tables for Inspection by attributes.

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

Federal Regulations:

29 CFR 110.1200 -Hazard Communications Standard

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Chemical society (ACS) Specifications:

Reagent Chemicals (Latest Edition)
Hydrochloric acid (HCl)
Toluene
n-Butanol

(Application for copies should be addressed to the American Chemical Society, 1122 sixteenth Street, N.W., Washington DC 20036.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials:

ASTM D13 -Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS.

3.1 **Materials and components.** Materials and components shall conform to the requirements specified herein.

3.1.1 **Bristle (Type I).** The bristle for Type I brushes shall be unbleached and undyed natural hog bristle. The bristle shall be boiled or steamed and straightened so that it will not twist or curl, and shall be free from reconditioned bristle or adulterants. The bristle shall pass the color test specified in 4.4.1 and the straightness test specified in 4.4.5. A certificate of compliance from the supplier may be accepted that the bristle meets the stiffness requirements.

3.1.2 **Synthetic filaments (Type II).** The synthetic filament for Type II brushes shall be an organic polymeric material conforming to H-S-951, class N, type I tapered filaments meeting the size designation of Table II. All filaments shall be tipped to a depth of not less than 3/16 inch (5 mm.). After tipping, 20 percent, by weight of the top-sized filaments, shall be flagged. (See 6.3.1)

3.1.3 **Ferrule.** The ferrule shall be made of corrosion-resistant metal, or metal made corrosion-resistant by a coating which shall not wrinkle, blister, separate, or become soft or dull when tested in accordance with 4.4.3.

3.1.4 **Handle.** The handle shall be either a smoothly finished close-grained hardwood, or a synthetic material with all sprues and runners removed. The handle shall not crack or break when tested as in 4.4.4.

3.1.5 **Filler strips.** Filler strips shall be made of wood, metal, or plastic and shall not soften or otherwise change when tested as in 4.4.3. Cardboard filler strips are allowed for grade B and C brushes.

3.1.6 **Setting Compound.** The setting compound shall be a material which shall withstand the test specified in 4.4.3 without softening or deteriorating, and without loosening of bristle or filament in the compound.

3.2 **Construction and physical requirements.**

3.2.1 **Ferrule.** The ferrule metal shall have a minimum wall thickness of 0.010 inch (3 mm.). The ferrule shall be formed with an interlocked, lock-seamed, soldered, or welded end lap, and shall have a groove at the base to aid in controlling the flow of the setting compound. The ferrule height shall be 1-1/4 inch (32 mm.) minimum for grade C brushes and 1-15/32 inch (37 mm.) minimum for grade AA, A and B brushes.

3.2.2 **Set.** The butt ends of the bristles or filaments shall be set in a setting compound conforming to 3.1.6. When tested, as in 4.4.3.3, there shall be no more than a total of five (5) bristles or filaments pulled loose during the tensile pull.

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3.2.3 **Assembly.** For grades AA and A brushes, the ferrule shall be fastened to the handle with 17 gage, corrosion-resistant nails, 3/8 inch (10 mm.) minimum length. There shall be at least two (2) nails on each side of brushes 3-inch (76 mm.) wide or smaller, three (3) nails on each side of the 3-1/2 inch (89 mm.) and 4-inch (102 mm.) brushes, and four (4) nails on each side of the 5-inch (127 mm.) and 6-inch (152 mm.) brushes. Nails shall be evenly spaced on each side of the brush.

3.2.4 Trim.

a) All grade AA brushes shall be formed to a square top with flagged ends preserved, or cup chiseled, with a minimum chisel of 1/4 inch (6 mm.).

b) Grade A, and grade B brushes ((3-inches (76 mm.) or wider)).and all grade C brushes shall be formed to a square top, with tipping and flagging preserved.

c) Three-inch (76 mm.) brushes shall be formed to a square top (with the tipped or flagged ends preserved), or cupped or machined to a chisel edge with a minimum chisel of 1/4 inch (6 mm.). (If cup chiseled, flag ends shall be preserved. If machine chiseled, the ends shall be sanded to a fine, soft edge.)

d) Grade A and grade B brushes ((less than 3-inches (76 mm.) wide)) shall be cupped or machined to a chisel edge with a minimum chisel of 1/4 inch (6 mm.). If cup chiseled, flagged ends shall be preserved. If machine chiseled, the ends shall be sanded to a fine, soft edge.

3.3 Composition and physical requirements.

3.3.1 **Type I.** The bristle shall be 60 percent top size and blended in accordance with Table I. (Dressed bristle, described as 60 percent top size, denotes that bundle consists of 60 percent bristle measuring 1/16 inch (2 mm.) or more above the length specified to 1/8 inch (3 mm.) below the length specified; 30 percent to consist of bristle from 1/8 (3 mm.) to 3/8 inch (10 mm.) below the length specified; 10 percent will be 3/8 (10 mm.) to not more than 1/2 inch (13 mm.) below specified length, but including a few unavoidable shorts.)

TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Percentage (wt.) & length of bristle in., (mm.), Min.	Brush thickness inside ferrule, in., (mm.),	Bristle length outside ferrule, inches, (mm.), Min.	Bristle weight per brush, gm, Min.
AA	6 (152) [square or cup]	80% 5-1/4 (133) 10% 5 (127) 10% 4-1/2 (114)	1 (25)	4-15/16 (125)	153

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TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Percentage (wt.) & length of bristle in., (mm.), Min.	Brush thickness inside ferrule, in., (mm.),	Bristle length outside ferrule, inches, (mm.), Min.	Bristle weight per brush, gm, Min.
AA	5 (12.70) [square or cup]	80% 5-1/4 (133) 10% 5 (127) 5% 4-3/4 (121) 5% 4-1/2 (114)	1 (25)	4-15/16 (125)	128
AA	4 (102) [square or cup]	80% 5 (127) 10% 4-3/4 (121) 5% 4-1/2 (114) 5% 4-1/4 (108)	1 (25)	4-11/16 (119)	100
AA	3-1/2 (89) [square or cup]	80% 4-3/4 (121) 10% 4-1/2 (114) 5% 4-1/4 (108) 5% 4 (102)	1 (25)	4-7/16 (113)	82
A	6 (152) [square]	80% 4-3/4 (121) 10% 4-1/2 (114) 5% 4-1/4 (108) 5% 4 (102)	1 (25)	4-7/16 (113)	125
A	5 (127) [square]	80% 4-3/4 (121) 10% 4-1/2 (114) 5% 4-1/4 (108) 5% 4 (102)	1 (25)	4-7/16 (114)	105
A	4 (102) [square]	80% 4-1/2 (114) 10% 4-1/4 (108) 5% 4 (102) 5% 3-3/4 (95)	1 (25)	4-3/16 (106)	79
A	3.5 (89) [square]	80% 4-1/2 (114) 10% 4 (102) 5% 3-3/4 (95) 5% 3-1/2 (89)	1 (25)	4-3/16 (106)	65
A	3 (76) [square]	80% 3-3/4 (95) 15% 3-1/2 (89) 5% 3-1/4 (83)	13/16 (21)	3-5/16 (84)	60
A	2.5 (64) [chisel]	80% 3-1/2 (89) 15% 3-1/4 (83) 5% 3 (76)	13/16 (21)	3-1/16 (78)	40
A	2 (51) [chisel]	80% 3-1/4 (83) 15% 3 (76) 5% 2-3/4 (70)	3/4 (19)	2-13/16 (71)	27
A	1.5 (38) [chisel]	80% 3 (76) 15% 2-3/4 (70) 5% 2-1/2 (64)	11/16 (18)	2-9/16 (65)	16
B	5 (127) [square]	80% 4-1/2 (114) 10% 4-1/4 (108) 5% 4 (102) 5% 3-3/4 (95)	7/8 (22)	4-3/16 (106)	96

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TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Percentage (wt.) & length of bristle in., (mm.), Min.	Brush thickness inside ferrule, in., (mm.),	Bristle length outside ferrule, inches, (mm.), Min.	Bristle weight per brush, gm, Min.
B	4 (102) [square]	80% 4 (102) 10% 3-3/4 (95) 5% 3-1/2 (89) 5% 3-1/4 (83)	7/8 (22)	3-11/16 (94)	68
B	3.5 (89) [square]	80% 3-3/4 (95) 10% 3-1/2 (89) 5% 3-1/4 (83) 5% 3 (76)	7/8 (22)	3-7/16 (87)	57
B	3 (76) [square]	80% 3-1/2 (89) 10% 3-1/4 (8.3) 5% 3 (76) 5% 2-3/4 (70)	11/16 (18)	3-3/16 (81)	45
B	3 (76) [chisel]	80% 3-1/4 (83) 15% 3 (76) 5% 2-3/4 (70)	11/16 (18)	2-3/16 (81)	40
B	2.5 (64) [chisel]	80% 3 (76) 15% 2-3/4 (70) 5% 2-1/2 (64)	5/8 (16)	2-9/16 (65)	27
B	2 (51) [chisel]	80% 2-3/4 (70) 20% 2-1/2 (64)	9/16 (14)	2-5/16 (59)	17
B	1.5 (38) [chisel]	100% 2-1/2 (64)	1/2 (13)	2-1/16 (52)	11
B	1 (25) [chisel]	100% 2-1/2 (64)	7/16 (11)	2-1/16 (52)	6
C	4 (102) [square]	80% 3-1/2 (89) 10% 3-1/4 (83) 5% 3 (76) 5% 2-3/4 (70)	7/8 (22)	3-3/16 (81)	54
C	3.5 (89) [square]	80% 3-1/4 (83) 10% 3 (76) 5% 3/4 (70) 5% 2-1/2 (64)	7/8 (22)	2-15/16 (75)	45
C	3 (76) [square]	80% 3 (76) 10% 2-3/4 (70) 5% 2-1/2 (64) 5% 2-1/4 (57)	11/16 (18)	2-11/16 (68)	37
C	3 (76) [square]	100% 2-3/4 (70)	1/2 (13)	2-3/16 (56)	28
C	2 (51)	100% 2 (51)	5/16 (8.0)	1-9/16 (4.0)	5
C	1.5 (4.0) [square]	100% 2-1/2 (64)	13/32 (10)	1-9/16 (4.0)	10
C	1 (25) [square]	100% 2 (51)	5/16 (8.0)	1-9/16 (4.0)	5

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TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Percentage (wt.) & length of bristle in., (mm.), Min.	Brush thickness inside ferrule, in., (mm.),	Bristle length outside ferrule, inches, (mm.), Min.	Bristle weight per brush, gm, Min.
C	0.5 (13) [square]	100% 2 (51)	1/4 (6)	1-9/16 (4)	2

Bristles 3-3/4 (95 mm.) through 3-1/2 inch (89 mm.) shall be 30% stiff, 30% medium, and 40% soft. Bristles 3-1/4 (83 mm.) through 3 inches (76 mm.) shall be 20% medium and 80% soft. Bristles shorter than 3 inches (76 mm.) shall be 100% soft. (A tolerance of plus or minus 3 percent is allowed for each length and stiffness specified.)

3.3.2 **Type II.** Synthetic filament conforming to 3.1.2 shall be blended as specified in table II, with a tolerance of plus or minus 3 percent for each length and stiffness specified.

TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, in., (mm.),Min. [Trim]	Filament size designation, percentage, and length inches, (mm.), Min.	Length of filament outside ferrule in.,(mm.),Min.	Brush thickness inside ferrule, in.,(mm)	Filament weight per brush gm., Min.
AA	5 (127) [square or cup]	**12-8:** 40% 4-3/4 (121) **15-10:** 15% 4-3/4 (121) 15% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102)	4-7/16 (113)	1(25)	130
AA	4 (102) [square or cup]	**12-8:** 40% 4-3/4 (121) **15-10:** 15% 4-3/4 (121) 15% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102)	4-7/16 (113)	1(25)	105
AA	3.5 (89) [square or cup]	**12-8:** 45% 4-1/2 (114) **15-10:** 10% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102) 15% 3-3/4 (95)	4-3/16 (106)	1(25)	91

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TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Filament size designation, percentage, and length inches, (mm.), Min.	Length of filament outside ferrule in., (mm.), Min.	Brush thickness inside ferrule, in., (mm)	Filament weight per brush gm., Min.
AA	3 (76) [square or cup]	**12-8:** 55% 4-1/2 (114) **15-10:** 15% 4 (102) 15% 3-3/4 (95) 15% 3-1/2 (89)	4-3/16 (106)	1 (25)	74
A	6 (152) [square]	**12-8:** 45% 4-1/2 (114) **15-10:** 10% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102) 15% 3-3/4 (95)	4-3/16 (106)	1(25)	145
A	5 (127) [square]	**12-8:** 45% 4-1/2 (114) **15-10:** 10% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102) 15% 3-3/4 (95)	4-3/16 (106)	1(25)	116
A	4 (102) [square]	**12-8:** 55% 4-1/4 (108) **15-10:** 15% 4 (102) 15% 3-3/4 (95) 15% 3-1/2 (89)	3-15/16 (100)	1 (25)	88
A	3.5 (89) [square]	**12-8:** 55% 4 (102) 15% 3-3/4 (95) **15-10:** 15% 3-1/2 (89) 15% 3-1/4 (83)	3-11/16 (94)	1(25)	77
A	3 (76) [square]	**12-8:** 55% 3-3/4 (95) 15% 3-1/2 (89) 15% 3-1/4 (83) **15-10:** 15% 3 (76)	3-7/16 (87)	1 (25)	68
A	3 (76) [chisel]	**12-8:** 20% 3-1/4 (83) 20% 3-1/2 (89) 40% 3-3/4 (95) **9-5:** 20% 3-3/4 (95)	3-5/16 (84)	13/16 (21)	54

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TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Filament size designation, percentage, and length inches, (mm.), Min.	Length of filament outside ferrule in., (mm.), Min.	Brush thickness inside ferrule, in., (mm)	Filament weight per brush gm., Min.
A	2.5 (64) [chisel]	**12-8:** 20% 3 (76) 20% 3-3/4 (95) 20% 3-1/2 (89) **9-5:** 40% 3-1/2 (89)	3-1/16 (78)	13/16 (21)	45
A	2 (51) [chisel]	**12-8:** 20% 2-3/4 (70) 20% 3 (76) 10% 3-1/4 (83) **9-5:** 50% 3-1/4 (83)	2-13/16 (71)	3/4 (19)	35
A	1.5 (38) [chisel]	**12-8:** 20% 2-1/2 (64) 20% 2-3/4 (70) **9-5:** 60% 3 (76)	2-9/16 (65)	11/16 (18)	18
B	5 (127) [square]	**12-8:** 45% 4-1/2 (114) **15-10:** 10% 4-1/2 (114) 15% 4-1/4 (108) 15% 4 (102) 15% 3-3/4 (95)	4-1/8 (105)	7/8 (22)	102
B	4 (102) [square]	**12-8:** 55% 4 (102) 15% 3-3/4 (95) **15-10:** 15% 3-1/2 (89) 15% 3-1/4 (83)	3-5/8 (92)	7/8 (22)	74
B	3.5 (89) [square]	**12-8:** 55% 3-3/4 (95) 15% 3-1/2 (89) 15% 3-1/4 (83) **15-10:** 15% 3 (76)	3-3/8 (86)	7/8 (22)	60
B	3 (76) [square]	**12-8:** 55% 3-1/2 (89) 15% 3-1/4 (83) 15% 3 (76) 15% 2-3/4 (70)	3-1/8 (79)	7/8 (22)	48

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Uniform to standards for sand, garnet, and gravel for mixed media filters.

TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Filament size designation, percentage, and length inches, (mm.), Min.	Length of filament outside ferrule in., (mm.), Min.	Brush thickness inside ferrule, in., (mm)	Filament weight per brush gm., Min.
B	3 (76) [chisel]	**12-8:** 20% 2-3/4 (70) 20% 3 (76) 10% 3-1/4 (83) **9-5:** 50% 3-1/4 (83)	2-3/16 (81)	11/16 (18)	42
B	2.5 (76) [chisel]	**12-8:** 20% 2-1/2 (64) 20% 2-3/4 (70) **9-5:** 60% 3 (76)	2-9/16 (65)	5/8 (16)	30
B	2 (51) [chisel]	**12-8:** 30% 2-1/2 (64) **9-5:** 70% 2-3/4 (70)	2-5/16 (59)	9/16 (14)	20
B	1.5 (38) [chisel]	**9-5:** 30% 2-1/4 (83) 70% 2-1/2 (64)	2-1/16 (52)	1/2 (13)	13
B	1 (25) [chisel]	**9-5:** 30% 2-1/4 (83) 70% 2-1/2 (64)	2-1/16 (52)	7/16 (11)	6
C	4 (102) [square]	**12-8:** 55% 3-1/2 (89) 15% 3-1/4 (83) 15% 3 (76) 15% 2-3/4 (70)	3-3/16 (81)	7/8 (22)	54
C	3.5 (89) [square]	**9-5:** 25% 3-1/4 (83) **12-8:** 30% 3-1/4 (83) 15% 3 (76) 15% 2-3/4 (70) 15% 2-1/2 (64)	2-15/16 (75)	7/8 (22)	45
C	3 (76) [square]	**9-5:** 40% 3 (76) **12-8:** 20% 3 (76) 20% 2-3/4 (70) 20% 2-1/2 (64)	2-11/16 (6.83)	7/8 (22)	34
C	3 (76) [square]	**9-5:** 100% 3 (76)	2-9/16 (65)	1/2 (13)	31
C	2.5 (64) [square]	**9-5:** 100% 2-3/4 (70)	2-5/16 (84)	15/32 (84)	21
C	2 (51) [square]	**9-5:** 100% 2-3/4 (70)	2-5/16 (84)	7/16 (11)	16

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TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, in., (mm.), Min. [Trim]	Filament size designation, percentage, and length inches, (mm.), Min.	Length of filament outside ferrule in., (mm.), Min.	Brush thickness inside ferrule, in., (mm)	Filament weight per brush gm., Min.
C	1.5 (38) [square]	9-5: 100% 2-1/2 (64)	2-1/16 (53)	13/32 (10)	10
C	1 (25) [square]	9-5: 100% 2-1/2 (64)	2-1/16 (52)	3/8 (9.5)	6
C	0.5 (13) [square]	9-5: 100% 2-1/2 (64)	1-9/16 (40)	1/4 (6.4)	2.75

3.4 Marking for identification. Unless otherwise specified (6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number (NSN).

3.5 Workmanship. The finished brushes shall be exempt of sharp edges, splinters, burrs, or other imperfections. In addition, the brushes shall be free of residual debris and other extraneous matter. The brushing part shall be trimmed evenly and uniformly with the flagged ends preserved to the finished length outside the ferrule.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 Inspection of preparation for delivery. An examination shall be made to determine compliance with 3.3 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 Inspection of the end item. Lots shall be assembled and randomly sampled in accordance with MIL-STD-105, inspection level II, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles or filaments
Ferrules
Handles
Construction
Set
Trim
Workmanship
Identification markings

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Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 Sampling for testing of the end item. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4. Testing procedures. Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 Color test. Samples of bristle shall be soaked in boiling HCl (ACS Reagent Grade, 5 percent by volume in distilled water) for 5 minutes. The solution shall show no color other than green. Any other color shall be indicative of coloring matter.

4.4.2 Weight of bristle or filament. After removal of the ferrule, the setting compound shall be removed from the bristle or filament by any practical means which does not injure the bristle or filament. The weight of bristles or filaments shall be determined for compliance with physical requirements.

4.4.3.1 Handle (Organic coating). Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a lacquer thinner conforming to CID A-A-857. After 30 minutes, the handle shall be removed from the thinner and inspected for wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a mineral spirits conforming to Federal Specification TT-T-291. After 30 minutes, the handle shall be removed from the mineral spirits and inspected for wrinkling, blistering, or separation of the coating.

C. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and inspected for wrinkling, blistering, or separation of the coating.

D. The brush handle shall be dried for 2 hours and the organic coating shall be examined for softening or dulling.

4.4.3.2 Base material, ferrule, filler strips, synthetic handles. The brush shall be tested for solvent resistance using a solvent composed of:

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Normal butyl acetate (Industrial grade)	15 +/- 1 percent by volume
Normal butyl alcohol (Industrial grade)	15 +/- 1 percent by volume
Methyl Ethyl Ketone (Industrial grade	10 +/- 1 percent by volume
Toluene (Industrial)	Remainder

(**NOTE:** The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, plating/coating on the ferrule, filler strips, and the setting compound.)

A. The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in the solvent mixture for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

B. **Handle and ferrule.** As soon as the brush is removed from the solvent, the organic coating on the ferrule shall be examined for wrinkling, blistering, or separation. The handle shall be examined for any deleterious effect by the solvent. The brush shall then be dried for 2 hours and re-examined for softening or dulling of the ferrule coating or deterioration of the handle base material.

C. **Filler strips.** When the brush is removed from the solvent, filler strips shall be examined for softening, swelling, or other deleterious change.

D. **Setting compound.** After removal of the brush from the solvent, the setting compound shall be examined for softening, loosening of bristle, and other solvent effects. After drying one hour, the brush shall be suspended by the handle ,in a holding device ,and a group of bristles, 1/4-inch (6 mm) in diameter, shall be given a tensile pull of ten (10) pounds (4.5kg) away from the ferrule. Repeat this tensile pull test, at random places, one time for each inch (25mm) of brush width. After thoroughly drying the brush, remove the ferrule and examine the setting compound for deterioration, bristle loosening, or other solvent damage.

4.4.4 Drop test. Remove bristle (or synthetic filament) and ferrule. Synthetic handles shall be dropped twice from a height of ten (10) feet onto a concrete surface. Grade B and C brushes, in which the handles are fastened by means other than nailing, shall be similarly dropped, as assembled, and examined for compliance with 3.1.4 and 3.2.3.

4.4.5 Straightness. Place assembled brush head so that the bristle is immersed in 60 degree C. water for five (5) minutes. Shake out excess water. Hang brush with bristle/filament down to dry completely.. Measure flare by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Disregard stray bristles/filaments.

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4.4.6 Tipping and flagging of filament. The depth of tipping and flagging shall be measured with a ruler and a 7-power magnifying glass.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (See 6.2).

6. NOTES

6.1 Intended use. Brushes made with natural hog bristle (Type I) are recommended for application of solvent (organic) borne paints and coatings. Brushes made with synthetic filaments (Type II) are recommended for water borne paints and coatings.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).
(d) Marking required (see 5.1).

6.3 Definitions.

6.3.1 Flagging. Flagging is a mechanical process to split filament ends to simulate the natural flag of hog bristle.

6.3.2 Tipping. Tipping is a process of grinding or sanding filament tips to produce a fine, soft, even tip.

PREPARING ACTIVITY

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITY

GSA-FSS
JUS-FPI

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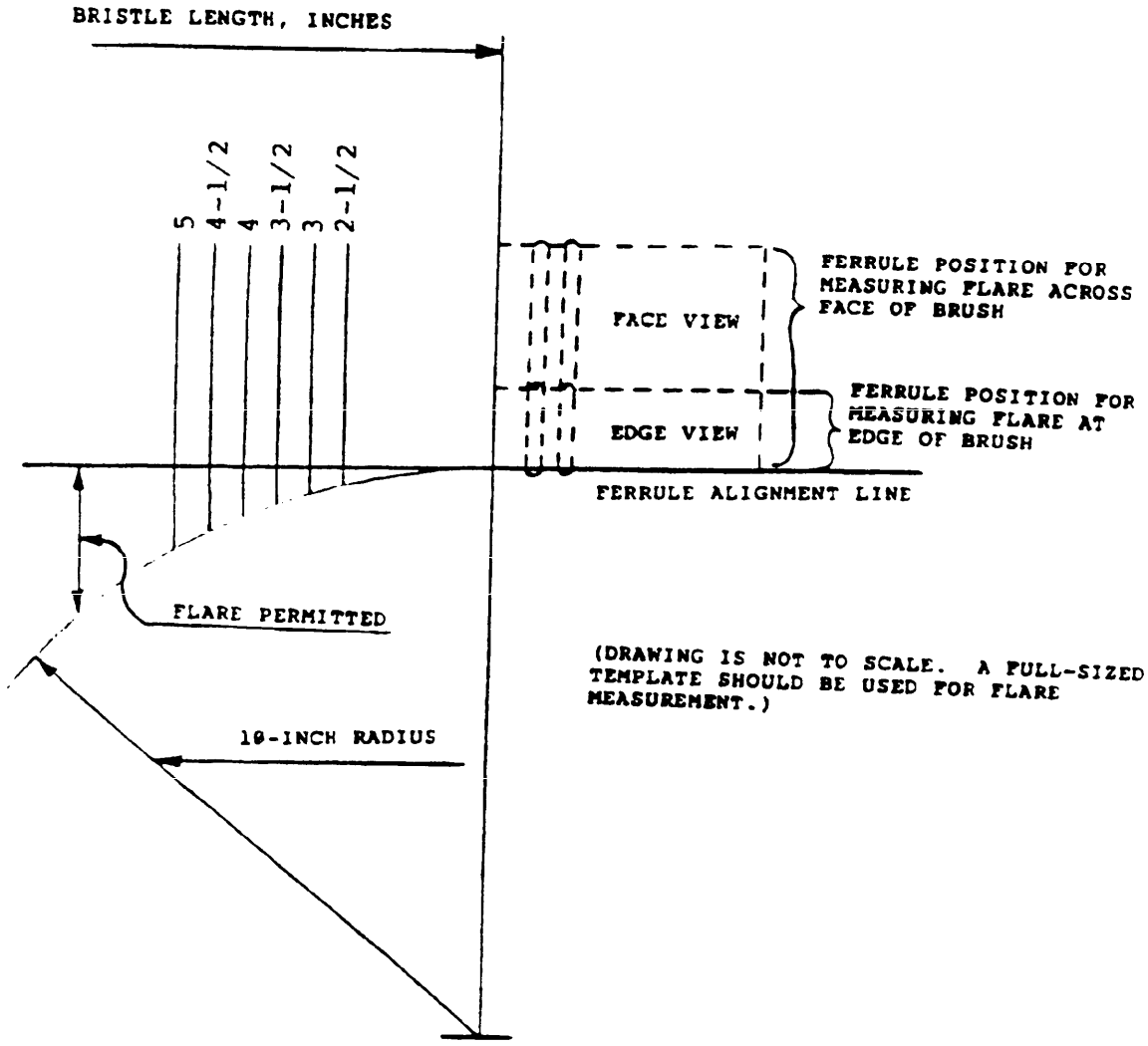


Figure 1. Flare measurement

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